Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/01/2005
PI Name:	Beckermann, Christoph Ph.D.		
Project Title:	Microstructure Evolution in Free Dendritic Growth		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	10/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNM04AA18G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The tasks of this project are: (1) to study transient growth of multiple interacting equiaxed dendrites; (2) to study free growth of binary alloy alloy dendrites in the presence of coupled heat and solute transport. The tasks will be accomplished through a combination of experiments, using transparent model substances in an existing dendrite growth setup, and numerical simulations using the phase-field and other methods. The objective of the project is to study microstructural evolution in free dendritic growth of pure substances and alloys from a supercooled melt. The research will not only close important gaps in our current understanding of dendritic growth, but is also needed for placing current simulations of grain structure evolution in castings on firmer theoretical grounds.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study is important for improving the understanding and computer simulation of solidification processing of metals and alloys.
Task Progress:	The following project tasks were completed during the past year: 1. Phase-field Simulations of Coupled Columnar and Equiaxed Dendritic Growth: Two-dimensional phase-field simulations of coupled columnar and equiaxed dendritic growth were performed for a binary alloy solidifying unidirectional under an imposed temperature gradient. The equiaxed grains nucleate and grow inside the constitutionally undercooled zone ahead of the columnar front. The phase-field model reduces to the sharp-interface equations in a thin-interface limit where kinetic and solute trapping effects are negligible. The computations are performed using an adaptive grid that moves with the solidification front. The competition between the columnar and equiaxed dendrites is investigated as a function of the nucleation undercooling, the equiaxed grain density, and the imposed cooling rate and temperature gradient. The conditions leading to a columnar-to-equiaxed transition are examined in detail and compared to theoretical models. 2. Transient Growth and Interaction of Equiaxed Dendrites: An improved version of a previously developed mesoscopic model was used to simulate transients and thermal interactions during growth of equiaxed dendrites of a pure substance. The model is validated through comparisons with exact, analytical solutions and direct, fully resolved phase-field simulations. The issue of constancy in the selection parameter, sigma*, during transients is addressed in some detail. The model is first applied to realistically simulate previously performed microgravity experiments involving the growth of succinonitrile dendrites from a stinger inside a growth chamber. It is shown how the thermal interactions between the seed and the dendrite and between the growth chamber wall and the dendrite cause temporal variations in the dendrite tip velocities. Excellent agreement with microgravity measurements is obtained. A scaling relation is derived that provides the duration of the seed size effect during the initial transient. The model is also used to investigate the transients arising during the growth of two equiaxed dendrites towards each other. A scaling relation for the duration of the transient decay of the tip velocities is derived. Additional study is needed to fully understand cases where equiaxed grains interact early before a fully dendritic structure is established. 3. Examination of Binary Alloy Free Dendritic Growth Theories with a Phase-Field Model: Two-dimensional phase-field simulations were used to test standard theories for free dendritic growth of alloys. While the transport of heat and solute in the melt is predicted well by the theories, the selection criterion for the operating state of the dendrite tip is found to break down in several respects. The selection parameter, sigma*, computed from the phase-field simulations varies strongly with alloy composition, Lewis number, and imposed undercooling, whereas the theories assume sigma* to be independent of these parameters. While the computed sigma* is the same for purely thermal and solutal dendrites, it experiences a minimum at a small but finite solute concentration where thermal and solutal effects are both important. A pronounced growth velocity maximum at this composition is therefore not found in the simulations. The high Peclet number corrections of the LKT theory are found to be ineffective for the present range of undercoolings. 4. Three-Dimensional Phase-Field Simulations of the Effect of Convection on Free Dendritic Growth: Three-dimensional free dendritic growth of a pure material into an undercooled melt in the presence of fluid flow was investigated numerically using the phase-field method. Such computations are made possible by solving the Navier–Stokes equations for the flow and the energy equation for the heat transport on a grid that is twice as coarse as the grid for the phase-field equation. The effect of the flow on the upstream growing dendrite tip velocity and radius of curvature is investigated as a function of the imposed flow velocity, undercooling, crystalline anisotropy, and Prandtl number. The results are compared to available theories of dendritic growth with and without convection. The predicted growth Peclet numbers as a function of the flow Peclet number are in reasonable agreement with the theoretical predictions. The dendrite tip selection parameter is essentially independent of the flow velocity within the range studied, which is also in accordance with theory. The three-dimensional dendrite tip shape is found to be well fitted by the same universal scaling relation as without flow. 5. Experiments on Free Dendritic Growth of Succinonitrile-Acetone Alloys: These experiments are well underway and results will be published in the coming project year.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Abstracts for Journals and Proceedings	Beckermann C. "Phase-Field Simulations of Coupled Columnar and Equiaxed Dendritic Growth." 2005 TMS Annual Meeting, San Francisco, CA, February 2005. 2005 TMS Annual Meeting, San Francisco, CA, February 2005. Invited Lecture. , Feb-2005
Articles in Peer-reviewed Journals	Steinbach I, Diepers H-J, Beckermann C. "Transient growth and interaction of equiaxed dendrites." Journal of Crystal Growth, 2005 Mar 1;275(3-4):624-38. http://dx.doi.org/10.1016/j.jcrysgro.2004.12.041 , Mar-2005
Articles in Peer-reviewed Journals	Ramirez JC, Beckermann C. "Examination of binary alloy free dendritic growth theories with a phase-field model." Acta Materialia, 2005 Apr;53(6):1721-36. http://dx.doi.org/10.1016/j.actamat.2004.12.021 , Apr-2005
Articles in Peer-reviewed Journals	Lu Y, Beckermann C, Ramirez JC. "Three-dimensional phase-field simulations of the effect of convection on free dendritic growth." Journal of Crystal Growth, 2005 Jun 15;2080(1-2):320-34. http://dx.doi.org/10.1016/j.jcrysgro.2005.03.063 , Jun-2005